From Rough to Radiant: Navigating the Diamond Price Landscape

Lab 2

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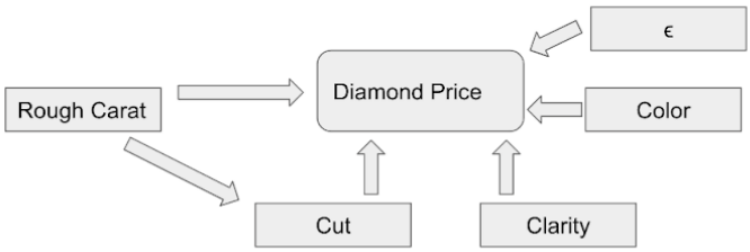
Introduction

In recent years, the diamond industry has experienced substantial growth and increased competition, resulting in higher demand for accurate pricing models that can account for the various factors influencing a diamond's value. This study aims to investigate the impact of the cut quality (X) of a diamond on its price (Y), using a regression analysis that controls for other factors (covariates) such as carat, color, and clarity. A successful predictive model could be instrumental for our company, DiamondOptics, which specializes in optimizing the diamond cutting process with advanced technology to maximize diamonds' values. By understanding the relationship between cut and price, DiamondOptics can offer tailored recommendations to diamond wholesalers, helping them prioritize investments in cutting technology and techniques that yield the greatest ROI. Overall, we want to construct a model that determines what kind of cut, based on diamond weight and other inherent qualities, would yield the highest revenue based on its estimated price. The initial causal path is as follows, where, as carat increases, cut quality and price also increases:



Data and Methodology

In this study, we use a publicly available Kaggle dataset[1] titled "Gemstone Price Prediction", containing ~27K observations on diamonds' characteristics such as carat, clarity, color, and cut quality. Missing values in the depth field were removed during exploratory data analysis, as depth cannot be imputed accurately. The 2015 dataset, priced in USD, was randomly split into training and test sets at a 70:30 ratio, respectively, using the 'sample' function, ensuring randomness.

Cause	Available Samples For Analysis (after removal for cause)	Remove number samples for cause
Start	26,967	
Missing Values from Depth	26,270	697
Training Data	18,389	7,881
Testing Data	7,881	18,389

Figure 1: Observation Accounting Table

To address the research question, a linear regression model was constructed to evaluate the relationship between specific design properties—namely cut quality—and the estimated price of the diamond. To operationalize the influence of the design property, cut, on price, we utilized the carat variable to identify the optimal cuts and weights of diamonds for maximizing the diamond's selling price. This approach allows us to determine the most profitable cut for a given diamond weight, providing actionable insights for our pricing analysts.

The operationalization of variables within the linear regression model is as follows: Treatment Variable includes the design property we are influencing "cut quality". Cut quality, represented by the "cut" column

in the dataset, includes categorical values ranked from "Fair", "Good", "Very Good", "Premium", and "Ideal." One-hot encoding was employed to create binary (dummy) variables for each cut category, with the "Ideal" cut serving as the reference. The Outcome Variable in the model is "price", indicating the estimated price of each diamond. Fig. 2 displays the spread of each diamond cut based on its weight and price.

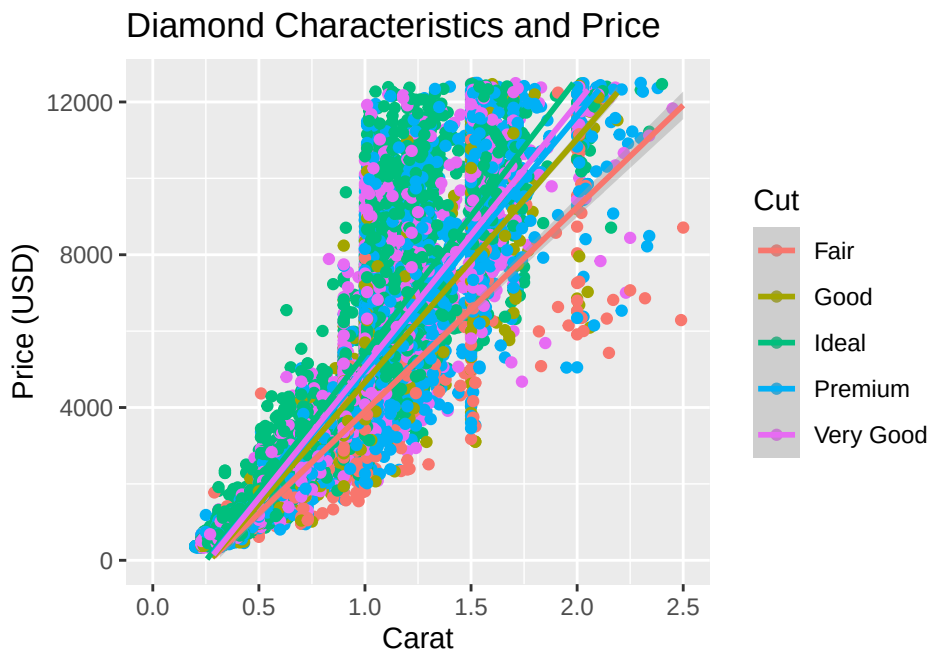


Figure 2: Diamond Characteristics

The linear regression model establishes the relationship between conceptual variables (Cut Quality and Price) and their operational definitions using specific column names ("cut" and "price") controlling for the other covariates. However, there are some key gaps between conceptual and operational definitions. Notably, the "cut" field in the dataset provides only five categories of cut quality, which may not capture the full spectrum of cut quality. Additionally, the one-hot encoding approach requires selecting a reference category, which could impact the interpretation of coefficients for other cut quality levels. Deciding to control for the covariates carat, color (diamond impurities), and clarity (absence of inclusions) was based on the research question and desire to build a model that directly addresses the relationship between cut and price. Although omitted variable bias may be a concern, all other covariates (depth & table) were left out of the model, as they added extra noise.

Model 1 demonstrates a linear regression on solely the treatment variable (cut) on the outcome variable (price). Model 2 uses the same dummy cut variables, using the "Ideal" cut as a reference level, with the addition of controlling for carat. Model 3 includes the other two control variables, clarity and color. Regression estimates on the test set are displayed in Table 1.

Results and Discussion

Our study used regression analysis to measure the impact of different diamond cut qualities on price compared to the "Ideal" cut. Our goal was to identify which cut could maximize profits. For instance, selling a "Very Good" cut diamond at a price similar to a "Premium" cut diamond of the same weight could increase profitability. The model's coefficients show the estimated price differences for each cut quality compared to the "Ideal" cut, and the p-values indicate the statistical significance of these relationships.

Table 1:

	Dependent variable:		
	Test Results of Coefficients - Linear Regression		
	(1)	(2)	(3)
Color			330.625*** (8.692)
Clarity			525.200*** (9.224)
Depth			−11.996 (10.443)
Carat		7,808.233*** (36.506)	8,751.639*** (33.100)
Cut = Fair	1,166.765*** (270.366)	−1,606.070*** (104.424)	−1,153.231*** (86.922)
Cut = Good	536.801*** (162.798)	−651.051*** (62.638)	−343.037*** (51.578)
Cut = Premium	1,121.394*** (113.614)	−371.257*** (44.098)	−135.517*** (36.303)
Cut = Very Good	498.360*** (117.694)	−299.173*** (45.260)	−135.023*** (36.822)
Constant	3,437.602*** (70.545)	−2,044.657*** (37.255)	−5,774.560*** (653.210)
Observations	7,881	7,881	7,881
R^2	0.013	0.855	0.905
Adjusted R^2	0.013	0.855	0.905
Residual Std. Error	3,966.866 (df = 7876)	1,520.283 (df = 7875)	1,230.505 (df = 7872)
Note:			*p<0.1; **p<0.05; ***p<0.01

In the third model, the estimated change in price for a one-unit increase in carat is ~$8,751, holding other variables constant. Diamonds with a "Fair" cut, on average, exhibited a price approximately $1,153 lower than diamonds with an "Ideal" cut, a finding deemed statistically significant (p-value ≈ 0). The "Good" cut, with a coefficient of -343.04 and a p-value below 0.01, revealed a statistically insignificant price decrease of $343.04 compared to the "Ideal" cut. Similarly, the "Premium" cut's coefficient of -135.52 and near-zero p-value suggested a statistically significant price decrease of $135.52. The "Very Good" cut's coefficient of -135.02 was also statistically significant. In order to further operationalize the results, an F-test was conducted, producing values over 1,000, determining practical significance in the context of our use case. Furthermore, the VIF of all variables in the model were close to one, signifying no perfect collinearity. Based on these estimated prices, a decision can be made to cut and sell "Very Good" cuts over "Premium" cuts due to the small difference in price. Consequently, the additional overhead costs involved in acquiring a premium cut can be avoided; thus, increasing profits.

Adding control variables improved model performance. This reinforces the concept that the four C's of a diamond – carat, cut, clarity, and color – collectively contribute to the price and overall value of the diamond (fig. 2). The first model's results are surprisingly statistically significant (p-values < 0.05), but the low R^2 value below one can explain the true relationship is not linear and the model has low explanatory power. Also, the positive coefficients likely signify correlation between the independent variables, meaning high collinearity without controlling for the other variables. The second model also showed statistically significant results, although controlling for clarity and color aided the model's predictive ability. The interpretation of coefficients and the decision to sell "Very Good" cuts from the third model holds for the second model as well.

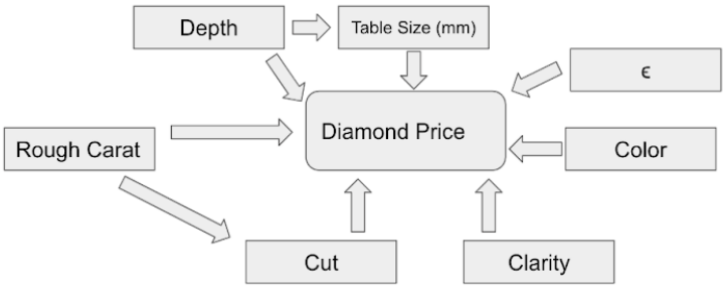
Limitations

Given the size of our dataset, we were able to rely on large sample model assumptions in the construction of each regression model. This includes the assumption that the observations in our analysis are independent and identically distributed (IID) and that a unique best linear predictor (BLP) of diamond prices exists. Because we know little about the diamonds' source, there is potential for clustering in the data given that some diamonds likely come from the same geographies and may share similar characteristics. Additionally, diamonds are often sold at standardized carat measures with similar cuts, which may further undermine the

IID assumption. To understand if a BLP exists for this dataset, it was important to confirm that a finite variance was present. Both carat sizes and price no doubt exhibit a heavy-tailed distribution and a roughly curvilinear relationship, which puts this assumption into doubt. Nonetheless, physical constraints suggest that the variance of the distribution of diamond characteristics is finite.

Additionally, to ascertain whether DiamondOptics' pricing model remains robust when employing a smaller sample size under the Classical Linear Model (CLM) assumptions, we have identified certain assumptions that may present challenges for our analysis. First, we assessed whether there was no perfect collinearity. We noted some relationships between predictor variables, while not perfectly collinear, do exhibit a linear relationship. For instance, carat (weight), volume, and depth maintain a strong linear relationship. Next, when including all variables in our model with minimal transformations (level model instead of log), the model does not perfectly satisfy the assumption of linear conditional expectation. Our third model exhibits some heteroskedasticity (non-constant error variance), suggesting that some exogenous variables that affect diamond prices may not be captured in our model. This would be a limitation in a model not relying on large sample assumptions. Finally, the assumption of normally distributed errors may not be perfectly satisfied by our dataset as the residuals are not symmetric around zero. Some of these CLM model assumptions have the potential to create issues if it were necessary to apply the conclusions on a smaller dataset.

There are a few omitted variables from our dataset that could bias the model results (with negative coefficients per Table 1). One being depth (the distance between the flat surface of the top of the diamond and pointed bottom). A diamond with depth has a positive relationship with both price and cut. Therefore, $\beta^r > \beta^c$, so the direction of the bias is moving towards zero. Another omitted variable is the diamond's "table" (width of the top of a diamond relative to the widest point). A diamond's table size has a positive relationship with price but a negative relationship with cut, as the ideal table depends on the cut. $\beta^r < \beta^c$, meaning the bias is moving away from zero in a negative direction. Below represents an updated causal diagram with these variables not previously considered.



Conclusion

Our data analysis shows that the rough carat influences cut size, which affects selling price. Our findings suggest that "Very Good" cuts, despite being lower on the scale than "Premium" cuts, may be more profitable by weight. This is an important finding as it implies that the cost to upgrade to "Premium" cuts may not always be profitable. Additionally, it can be inferred that there's an optimal trade-off between diamond price, carat, and cut. In future research, we would seek to investigate the costs of each diamond cutting upgrade to optimize our predictive model for maximum profit. Additional feature engineering and outcome variable estimation could improve the model, such as estimating price per carat (rather than solely price) and using a logarithmic scale. Nonetheless, our current model suffices for our intended use case. Another area we would like to understand in future analysis involves the potential impact of exogenous diamond industry variables (supply, demand, and mining costs) and the effect of wider economic conditions on diamond price. Although this was outside the scope of our current model, further research may enable us to account for these variables and improve our model's predictive ability.

Sources

1. Co-Learning Lounge (2015, August). Gemstone Price Prediction. Retrieved April 18, 2023 from https://www.kaggle.com/datasets/colearninglounge/gemstone-price-prediction?resource=download&select=cubic_zirconia.csv.

2. Why aren't all diamonds cut to ideal proportions. Beyond 4Cs. (n.d.). Retrieved April 18, 2023, from https://beyond4cs.com/shapes/round/why-diamonds-are-not-cut-ideally/